Anomalie and Outliers in TS

WARM UP QUESTIONS 1/1

- A monthly demand series has a clear upward trend and repeating yearly peaks. Which model features would you consider before forecasting it with ARIMA, and why?
- Are ARIMA and ETS linear or nonlinear models, and why?

Why?

Detecting outliers and anomalies and replacing or discarding them before fitting a forecasting model can positively increase the forecaster's performance.

How?

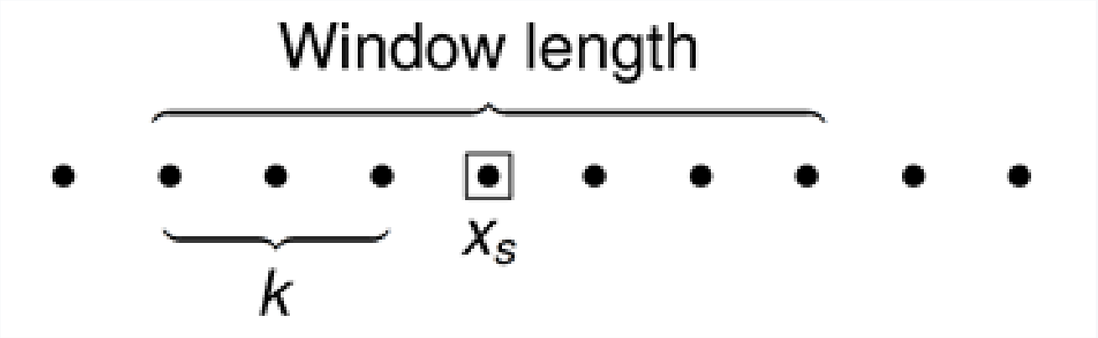
Standard outlier filters can be adapted to time series data by applying them to **sliding windows** of the original signal

Sigma Filter and Hample filter

Sigma filter: mark x_t as outlier if $|x - \bar{x}| > 3\sigma$. Under normal distribution of the data, this filter corresponds to keep 99.7% of the observations

The Hample filter works by first calculating the mean and standard deviation of a sliding window of data points.

1. Centers the window of odd length at the current sample x_s.
2. Computes the local median, $\bar{x}_s$, and standard deviation, σ_s, over the current window of data.
3. Compares the current sample with $n_\sigma \sigma_s$, where n_σ is the threshold value. If $|x_s - \bar{x}_s| > n_\sigma \sigma_s$, the filter identifies the current sample, x_s, as an outlier and replaces it with the median value, $\bar{x}_s$.



Boxplot Filter

The boxplot filter doesn't assume a particular distribution. An observation is kept if:

$$q_{0.25} - 1.5IQR \leq x \leq q_{0.25} + 1.5IQR$$

where IQR is the interquartile range = $q_{0.75} - q_{0.25}$. We can replace the sigma rule with the boxplot rule in the Hample filter to adapt it to time series data.

❓ Do you think these filters can be combined with STL decomposition for seasonal time series? How?

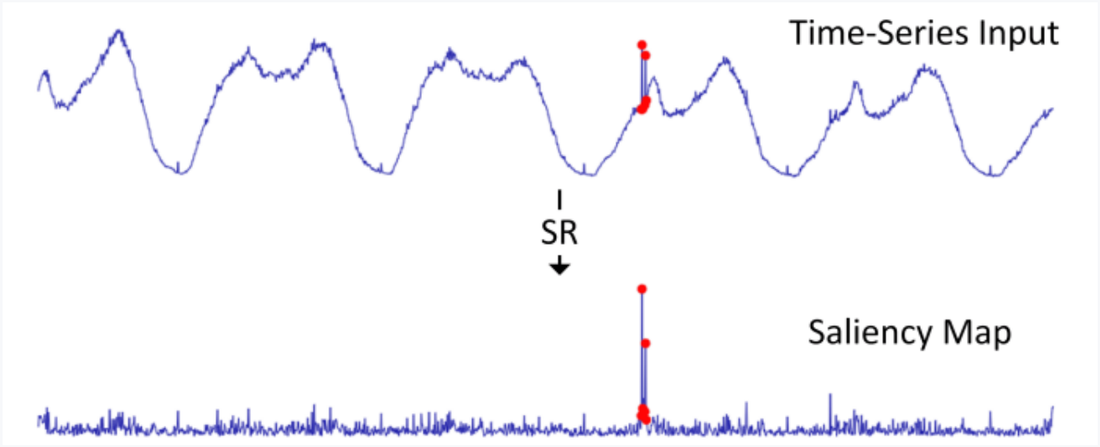
Spectral Residual Technique

SR technique was proposed by researchers at Microsoft to spot anomalies in time series. It exploits information in the frequency domain by taking the Fourier transform of the time series, and consists of the following steps:

1. Take the log of the absolute value of the Fourier transform of the time series.
2. Smooth the log values using a **rolling average**.
3. Compute the spectral residuals by subtracting the smoothed log values from the log values.
4. Inverse Fourier transform on the spectral residuals to obtain a **saliency map** or anomaly score

The anomaly score can then be used to identify regions of the time series that contain unusual activity. The SR technique has the advantage of being computationally efficient and easy to implement. Requires some parameter

$$A(f), P(f) = \text{ Amplitude } (\mathfrak{F}(\mathbf{x})), \text{ Phase } (\mathfrak{F}(\mathbf{x}))$$
$$L(f) = \log(A(f))$$
$$AL(f) = h_q(f) \cdot L(f)$$
$$R(f) = L(f) - AL(f)$$
$$S(\mathbf{x}) = \left\| \mathfrak{F}^{-1}(\exp(R(f) + iP(f))) \right\|$$

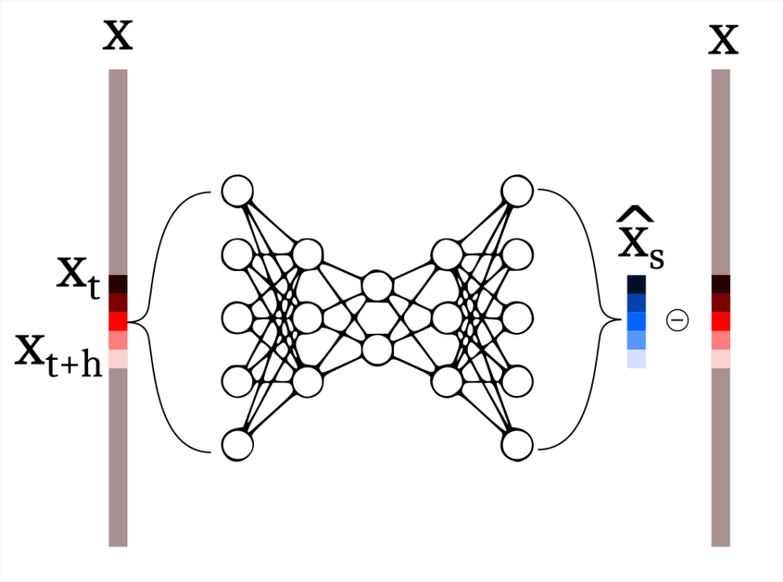


Time-Series Anomaly Detection Service at Microsoft, H.Ren et al, 2019

Autoencoders

Autoencoders can be used to generate **saliency maps** and detect anomalies in TS.

1. Train an autoencoder on "good" data to predict sliding windows of the original signal, $\hat{x}_s = [\hat{x}_t, \hat{x}_{t+1} \dots \hat{x}_{t+h}]$. The training is done minimizing reconstruction error $\|x_s - \hat{x}_s\|^2$
2. At inference time, run the autoencoder on newly observed data, and use the reconstruction error $\|x_s - \hat{x}_s\|^2$ as saliency map
3. Mark the observations as outliers if the reconstruction error is above a threshold



Schematics of a feed-forward autoencoder. The original sliding window is encoded in a vector with (usually) smaller dimensionality (latent representation), which is then used to reconstruct the original time series.

Forecast-based

We can combine ideas from the previous slides and define an anomaly detection strategy that is based on a forecasting model.

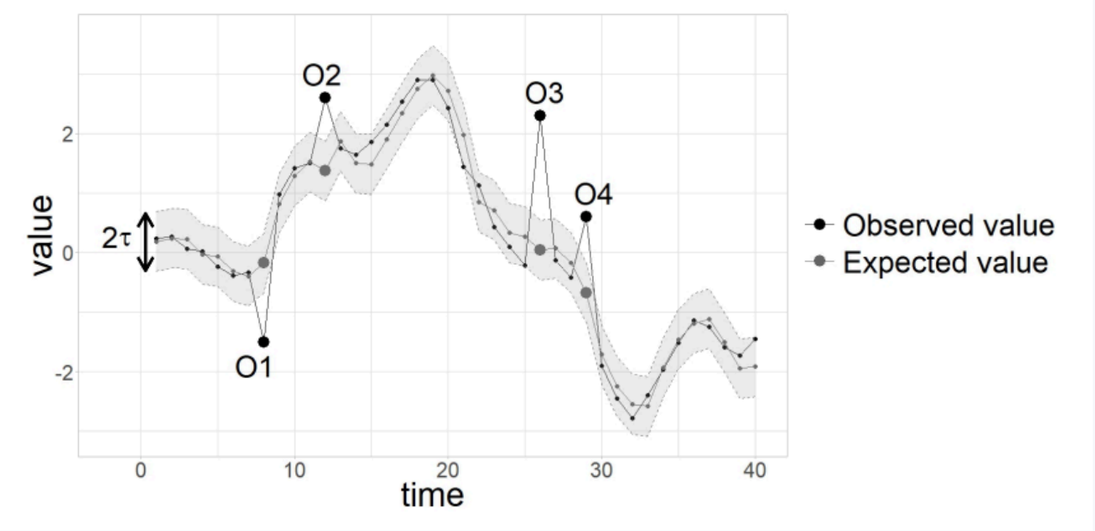
Given a forecasting model $f(x_t)$ we can write:

$$y_{t+k} = f(x_t) + \epsilon = \hat{y}_{t+k} + \epsilon$$

If the future point can show too extreme values of ϵ, we can mark it as an outlier.

$$|y_{t+k} - \hat{y}_{t+k}| > \tau$$

Usually, τ s chosen based on the assumed distribution of the error, for example, normal Gaussian. In this case, we can use 3σ for example.

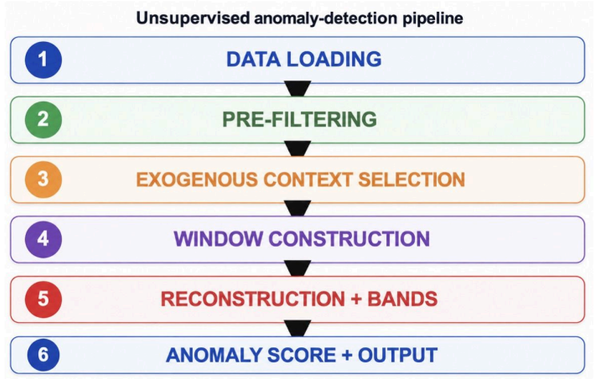


from: Blazquez-Garcia et al., A review on outlier/anomaly detection in time series data,2020 ACM

Anomaly detection for OASI - the problem

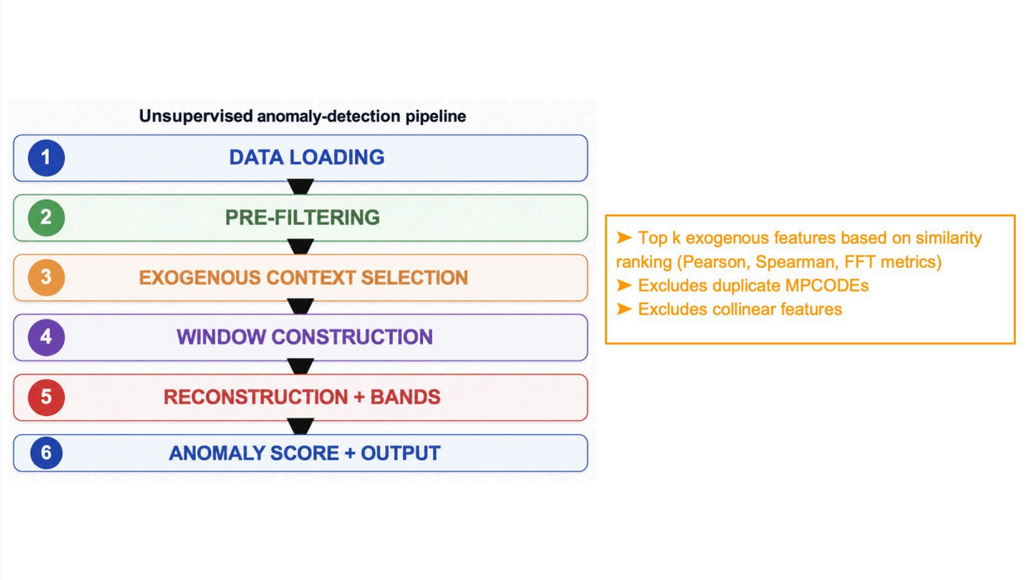
- OASI (Osservatorio Ambientale della Svizzera Italiana) wanted to update their automated anomaly detection pipeline
- No usable labels defining what an anomaly is
- Requirements
- must run on full catalogue (thousands of signals)
- must be multivariate (if x is high while y is low and usually it’s not, that’s an anomaly)
- must be explainable

Anomaly detection for OASI - the solution - 1

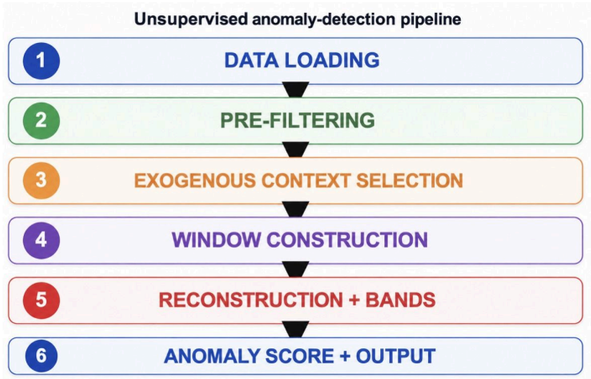


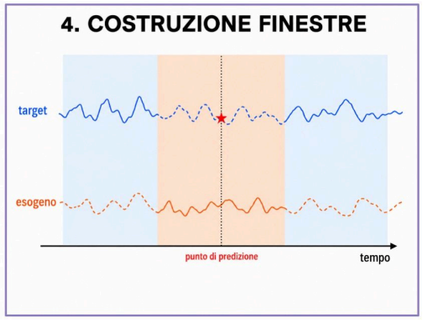
➤ Distribution filter applied to the individual time series
(detrended) + sigma filter at maximum temporal resolution.
➤ Outliers → removed & interpolated

Anomaly detection for OASI - the solution - 2

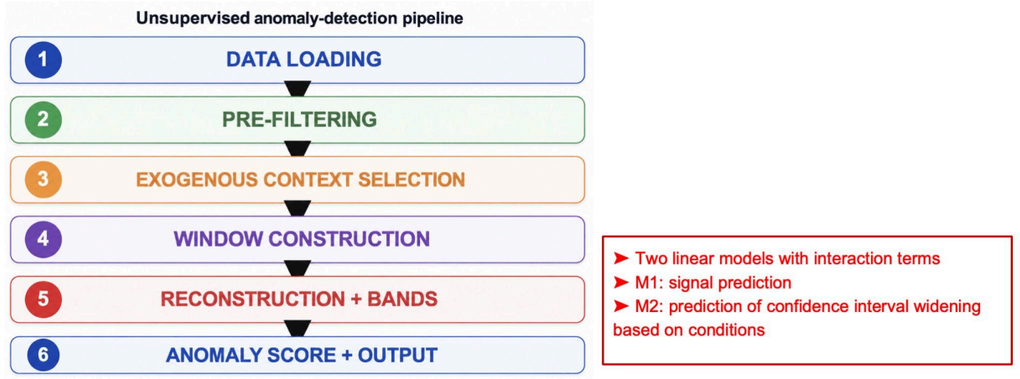


Anomaly detection for OASI - the solution - 3

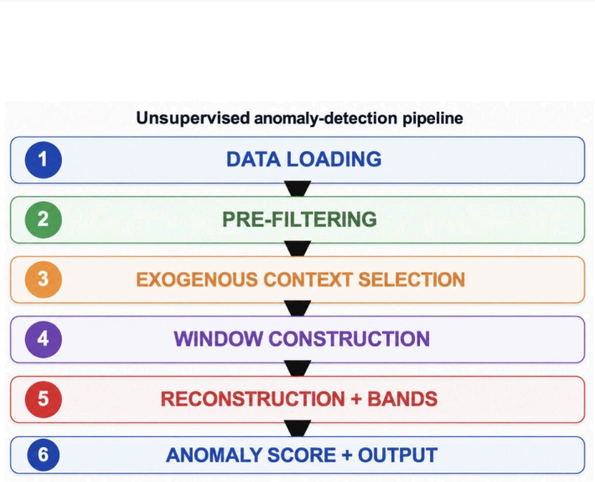


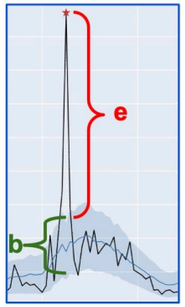


Anomaly detection for OASI - the solution - 4



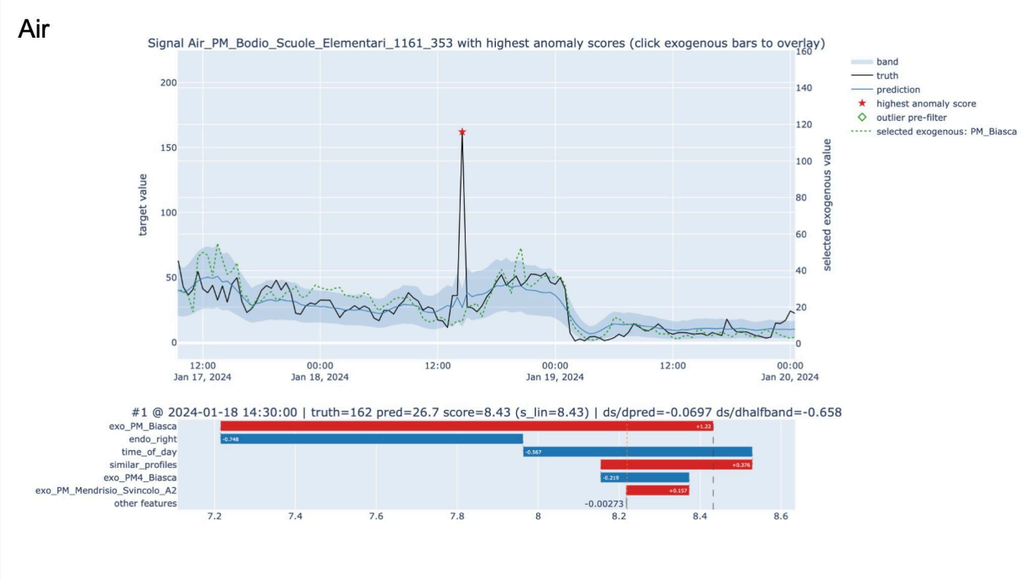
Anomaly detection for OASI - the solution - 5



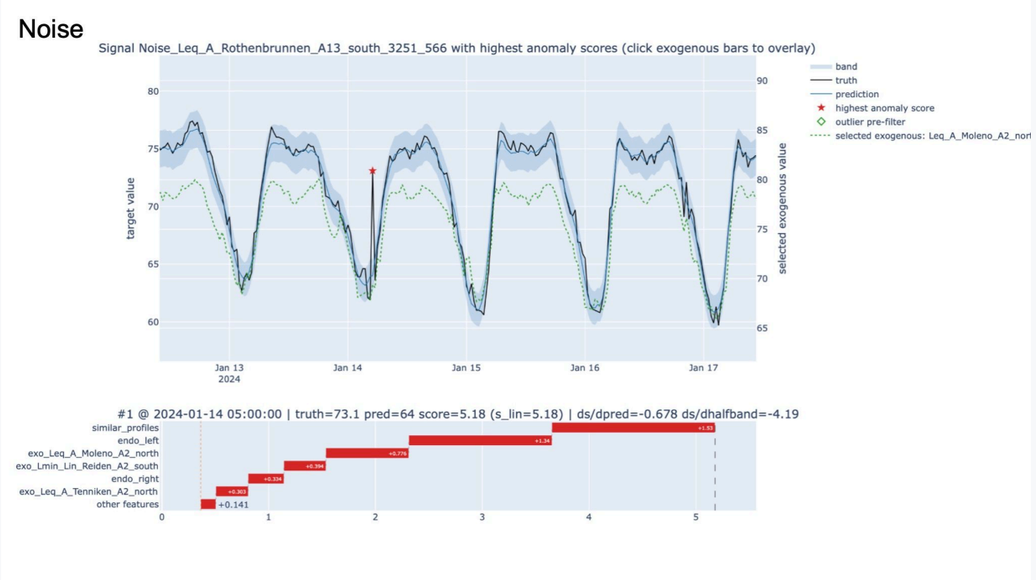


$S = e / b$

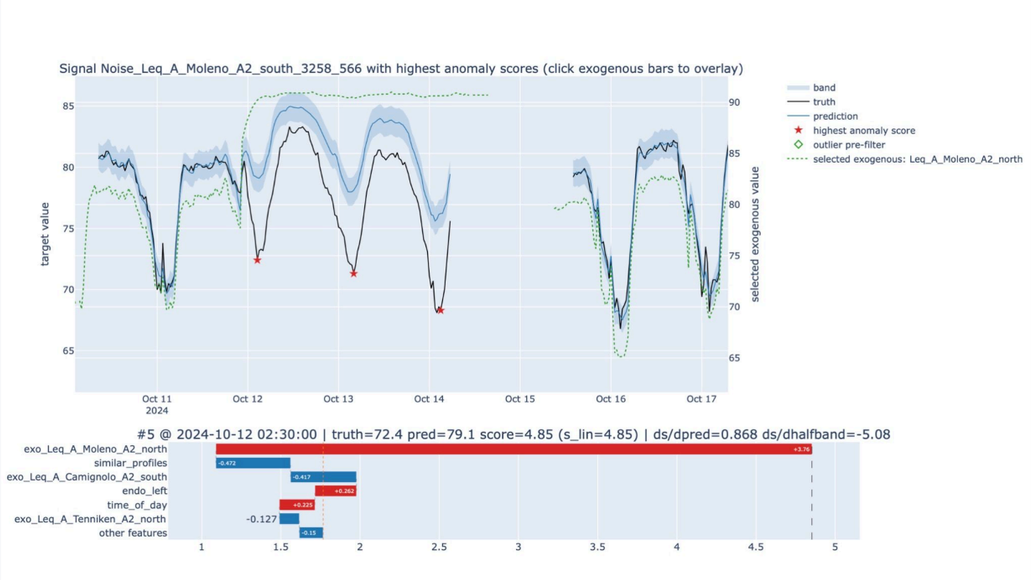
Anomaly detection for OASI - the solution - 6



Anomaly detection for OASI - the solution - 7



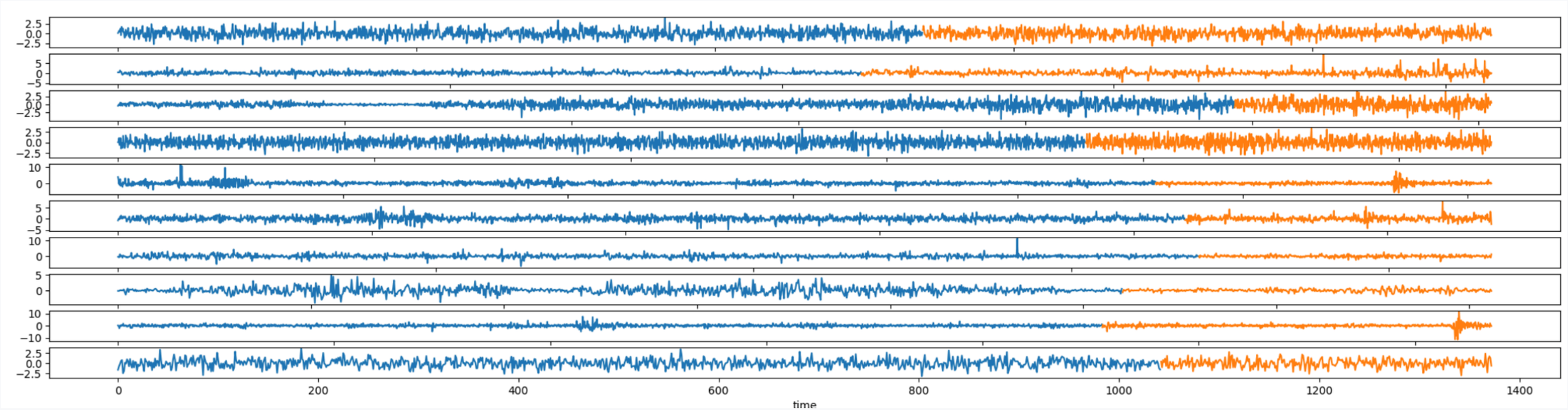
Anomaly detection for OASI - the solution - 8



AIDA Breakpoint detection - the problem

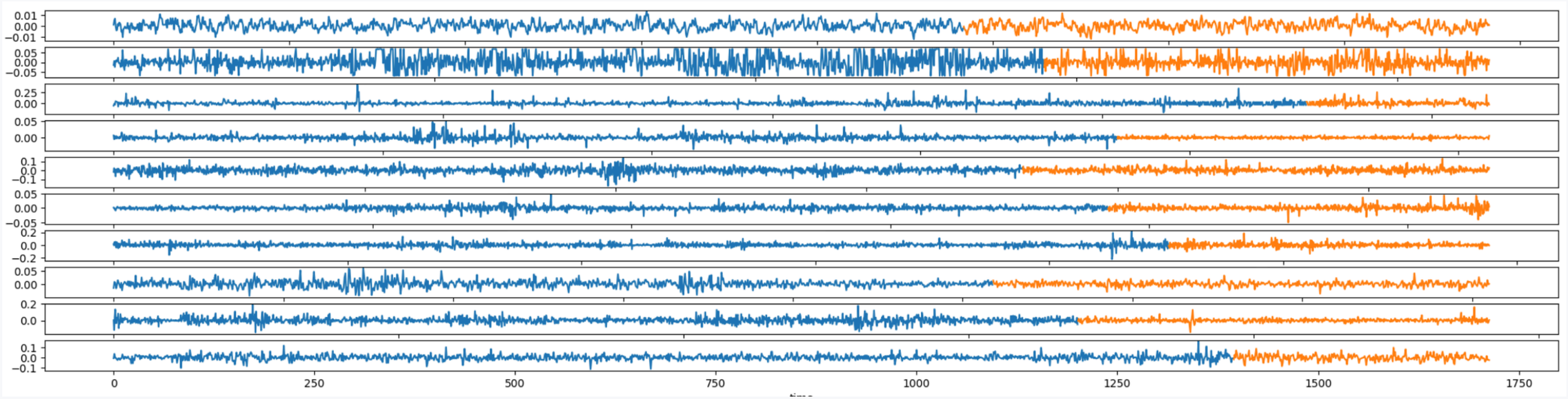
- The challenge asks you to detect whether a **known boundary point** splits a univariate time series into two genuinely different regimes.
- Given the full series—before and after the boundary—you must return a score from **0** (“no structural break”) to **1** (“break very likely”). The model is judged by ROC-AUC across unseen series, including changes in mean, volatility, dependence, distribution shape, or tails.

AIDA Breakpoint detection - Negative examples



negative examples - no breakpoint

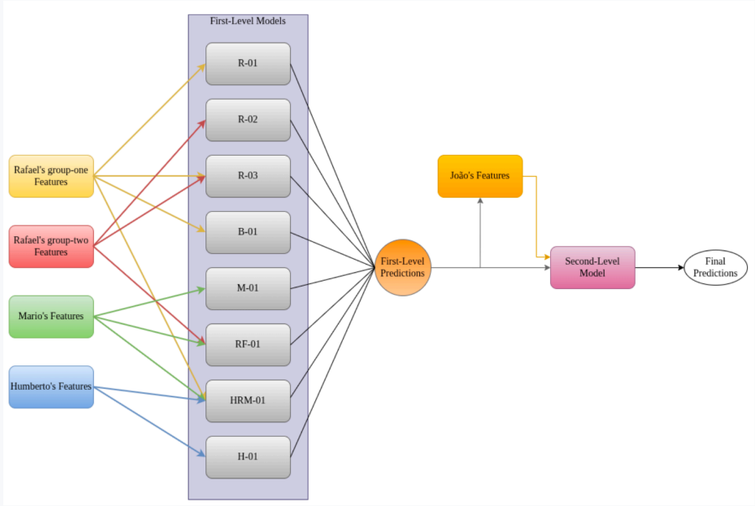
AIDA Breakpoint detection - Positive examples



positive examples - breakpoint

AIDA Breakpoint detection - the solution - 1

The team used a deliberately diverse **stacking** strategy: separate feature pipelines captured different types of pre/post-boundary change, then eight tree-based base models made independent predictions. Those predictions were combined with additional meta-features in a second-level model to produce the final structural-break score.



The solution of winning team

AIDA Breakpoint detection - the solution - 2

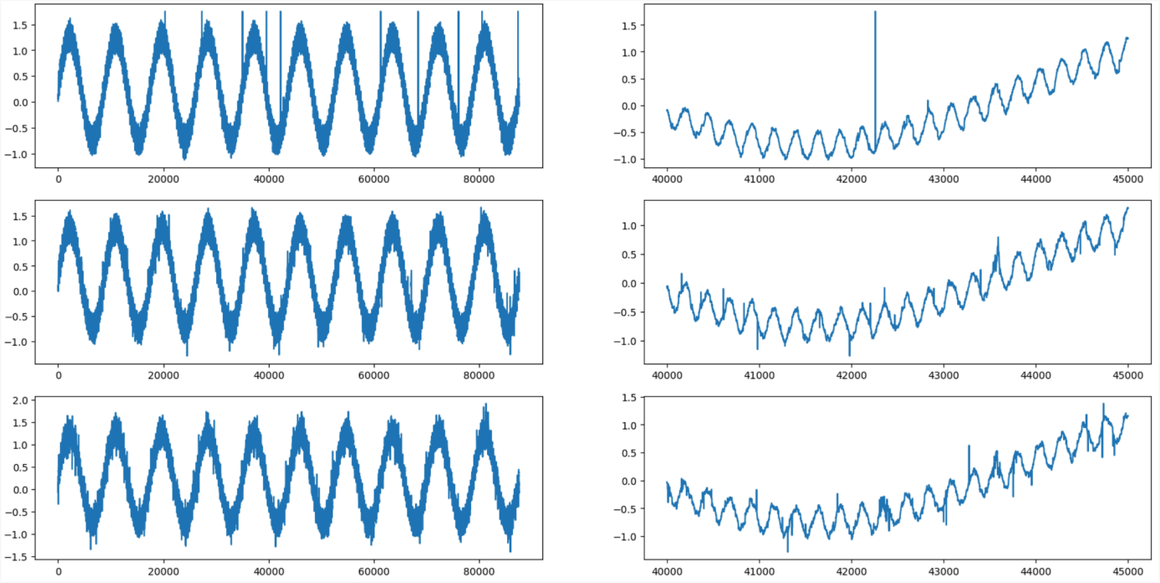
Among the indicators used

- diff entropy, sperman correlation, KS statistics, ACF
- Outlier filtering
- rolling windows stats

The competition

You got 3 time series with some anomalies

- Pointwise anomalies
- Structured anomalies (change of variance)



the time series of the competition

The competition

- Form a group
- Return a 3-columns CSV with binary values: 1=anomaly 0=regular point
- You'll be scored on $F1 = 2/(1/precision + 1/recall)$

$precision = TP/\hat{P}recall = TP/P$

The prize



I - 40 CHF

II - 10 CHF



Screenshot 2026-10-12 at 11.50.46.png